

Symmetric & Asymmetric Algorithms

The Symmetric Workhorses

Algorithm	Key	Status
AES-128/192/256	128-256 bits	Industry standard (FIPS)
ChaCha20	256 bits	Modern (TLS 1.3 default cipher; fast on phones)
DES / 3DES	56 / 112	Broken – never use

AES modes matter:

- ECB – insecure (same plaintext block → same ciphertext block; leaks patterns)
- CBC – better, but needs random IV + padding
- GCM / ChaCha20-Poly1305 – AEAD (Authenticated Encryption with Associated Data): encrypt + authenticate together

Rule: prefer AEAD (AES-GCM, ChaCha20-Poly1305). Non-AEAD modes let an attacker tamper with ciphertext.

The Asymmetric Algorithms

Algorithm	Purpose	Key size
RSA	Encrypt / sign	2048+ (3072 recommended)
ECC (ECDSA, ECDH)	Sign / key exchange	256-bit (≈ RSA 3072 strength)
DSA	Sign	Legacy, avoid

ECC is the modern choice – smaller keys, faster, same security (256-bit ECC ≈ 3072-bit RSA).

The Numbers Game: "Bits of Security"

128-bit security = industry standard (AES-128, ECC-256)

256-bit = also common for AES against quantum timeline debates

Think of "bits" as the $\log_2$ of the search space

A 56-bit DES key = ~72 quadrillion guesses. Feasible with GPUs. A 128-bit key is astronomically larger – effectively immune to brute force.

Post-Quantum (2026 Reality Check)

Quantum computers threaten RSA/ECC (Shor's algorithm), NOT symmetric badly (Grover only halves effective key length). Standards (NIST PQC / ML-KEM, ML-DSA) are emerging. Strategy now:

Use AES-256 symmetric (halves → 128 effective, still safe)

Just-safer: plan transition to PQC keys for the asymmetric parts

Putting It Together: A Full Session

Key exchange: ECDHE (Ephemeral) → session keys, forward secrecy

Bulk encryption: AES-256-GCM or ChaCha20-Poly1305

Authentication: server cert signed by CA → trust

That's exactly the TLS 1.3 handshake you saw in earlier lessons, now named properly.

Cipher Suites (Reading a TLS Config)

`TLS_AES_256_GCM_SHA384` = TLS 1.3 + AES-256 + GCM mode + SHA-384 for MAC.

Being able to decode a cipher suite string and judge whether it's modern is a real analyst skill.

← Hashing, MACs & Data Integri...

Digital Certificates & PKI →